Skills Summary

Ratios and Equivalent Ratio Tables

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a ratio off a model

Three ways to write one comparison:

a to b $\quad$ $a : b$ $\quad$ $\dfrac{a}{b}$

Order carries meaning. Whatever the question names first is written first. Count the model, write the pair, then simplify by dividing both numbers by a common factor — the model itself does not change, only the notation.

Example: A strip shows 8 star counters and 12 moon counters. Write stars to moons in simplest form.
Step 1. The question names stars first, so stars is the first number — this is a part compared with another part, read straight off the model.
Step 2. Stars to moons is 8 to 12, or 8 : 12, or $\frac{8}{12}$; dividing both by 4 gives $\mathbf{\frac{2}{3}}$ — for every 2 stars there are 3 moons.

Try it: A strip shows 9 triangle counters and 6 square counters. Write triangles to squares in simplest form.

check: $\mathbf{\frac{3}{2}}$

Skill 2 — Part-to-part or part-to-whole?

Rule:

If two parts are in the ratio $a : b$, the whole is $a + b$ equal shares.

part to part $= \dfrac{a}{b}$ $\qquad$ part to whole $= \dfrac{a}{a+b}$

Only the **part-to-whole** form may be read as "that fraction of the group." The denominator you need is usually a number the question never printed.

Example: A jar holds green marbles and blue marbles in the ratio 2 to 7. What fraction of the marbles is green?
Step 1. The question asks for a fraction *of the whole jar*, so the 7 is the wrong denominator — build the whole first.
Step 2. The whole is $2 + 7 = 9$ shares and green is 2 of them, so the fraction is $\mathbf{\frac{2}{9}}$.

Try it: A squad has forwards and defenders in the ratio 5 to 3. What fraction of the squad is forwards?

check: $\mathbf{\frac{5}{8}}$

Skill 3 — Equivalent ratio tables

Rule:

Multiply or divide **both** quantities by the same number to get an equivalent ratio. Every column of a ratio table names the same relationship, so any two columns may be set equal as a proportion and cross-multiplied.

Dividing down to a small, friendly column first usually makes the scale-up easy.

Example: 6 pens cost 15 dollars. What do 8 pens cost?
Step 1. The price per pen is steady, so the two columns are equivalent ratios — write a proportion rather than adding on 2 more dollars.
Step 2. $\frac{x}{8} = \frac{15}{6}$, so $6x = 120$ and the cost is **20** dollars. Check: 2 pens cost 5 dollars, and $4 \times 5 = 20$.

Try it: 4 notebooks cost 10 dollars. What do 14 notebooks cost?

check: **35**

Skill 4 — The value of a ratio

Rule:

The **value** of the ratio $a : b$ is the quotient $a \div b$ — a single number, the unit rate "a-units for every 1 b-unit."

Two ratios are equivalent exactly when their values are equal, so comparing values is how you decide which of two ratios is larger.

Example: A typist types 96 words in 8 minutes. Write the ratio of words to minutes and find its value.
Step 1. "Per minute" means the minutes must become 1, so divide the first quantity by the second.
Step 2. The ratio is $\frac{96}{8}$, and its value is $96 \div 8 = \mathbf{12}$ words per minute.

Try it: A craft kit puts 105 beads on 7 strings, the same number on each. Find the value of the ratio of beads to strings.

check: **15**

(!) Watch out: $a : b$ and $\frac{a}{b}$ are the same ratio, but "$\frac{a}{b}$ of the group" is a different claim. Before writing a fraction, ask: compared with the *other part*, or with the *whole*? If it is the whole, add the parts first.